

License Plate Detection via Transfer Learning with YOLOv5

Lucian Duncker (lduncker@wpi.edu) Joshua Pasino (jjpisano@wpi.edu)

Worcester Polytechnic Institute

ECE/CS 545 — Prof. Ziming Zhang

Abstract

We study automatic license plate detection in road-scene images. The task is straightforward to state: given an image, detect whether a license plate is present, and if so, draw a bounding box around it. We achieve this by fine-tuning YOLOv5s [3], pretrained on MS-COCO, into a single-class plate detector. Training used the public Kaggle license-plate dataset (approximately 27 000 annotated images) at 640×640 resolution for 20 epochs. Because YOLOv5s already provides strong defaults, minimal hyperparameter tuning was needed beyond selecting the Adam optimizer and keeping the standard augmentation schedule. Qualitative evaluation on held-out images shows confident, tightly localized predictions ($\text{conf} \geq 0.9$) on well-lit plates, with remaining false positives easily suppressed by a modest confidence threshold increase. The project demonstrates that off-the-shelf transfer learning on a general-purpose detector is sufficient for a practical plate-detection pipeline.

1. Motivation

We were interested in building an automatic license plate detector motivated by a concrete goal: running the model on Lucian’s dashcam footage to see how well a commodity detector handles real, unscripted highway driving.

2. Related Work

YOLO family. The YOLO [9] line frames detection as a single dense-grid regression. YOLOv2–v3 [7, 8] added anchor boxes and multi-scale heads; YOLOv4 [1] added Mosaic augmentation and CIoU loss. YOLOv5 [3], our choice, combines a CSPDarknet backbone with a PANet neck [5] and three detection heads at strides 8, 16, and 32.

Alternative architectures. SSD [6] and RetinaNet [4] are competitive one-stage detectors. Faster R-CNN [10] and DETR [2] offer stronger accuracy baselines at higher inference cost. EfficientDet [12] jointly scales backbone and neck for a range of compute budgets.

Plate-specific methods. WPOD-NET [11] predicts a per-detection affine warp to rectify each plate into a canonical view. Our work deliberately uses a general-purpose detector to measure how well commodity transfer learning performs without plate-specific inductive bias.

3. Technical Approach

3.1. Dataset

3.2. Model

3.3. Loss

3.4. Training

3.5. Inference

4. Results

Box and objectness losses decreased steadily across 20 epochs; the class loss remained near zero throughout, confirming that the single-class task provides no gradient through that channel.

Qualitative results on five held-out validation images are shown in ??.

The training data showed promising results with strong, confident detections. However, evaluation on real-world dashcam footage revealed performance gaps that could stem from several factors including image resolution, lighting conditions, and domain shift between the controlled dataset and unscripted highway driving.

4.1. Dashcam Video Evaluation

We evaluated the trained detector on dashcam clips recorded during real highway driving. All detection results are shown in ??.

4.2. Conclusion

Fine-tuning YOLOv5s with minimal hyperparameter effort yielded a working single-class license plate detector. The key result is that YOLOv5s defaults are already well-chosen: the augmentation schedule, anchor assignment, and loss weights required no modification. The partial-transfer

approach — warm-starting the backbone from COCO and training the detection heads from scratch — proved sufficient for the task without any plate-specific inductive bias.

The main remaining failure mode is low-confidence false positives on rectangular signage, which a simple confidence-threshold increase eliminates in practice.

Dataset Images



Dashcam Images



References

- [1] Alexey Bochkovskiy, Chien-Yao Wang, and Hong-Yuan Mark Liao. Yolov4: Optimal speed and accuracy of object detection. *arXiv preprint arXiv:2004.10934*, 2020.
- [2] Nicolas Carion, Francisco Massa, Gabriel Synnaeve, Nicolas Usunier, Alexander Kirillov, and Sergey Zagoruyko. End-to-end object detection with transformers. In *European Conference on Computer Vision (ECCV)*, pages 213–229, 2020.
- [3] Glenn Jocher. YOLOv5 by Ultralytics. <https://github.com/ultralytics/yolov5>, 2020.
- [4] Tsung-Yi Lin, Priya Goyal, Ross Girshick, Kaiming He, and Piotr Dollár. Focal loss for dense object detection. In *Proceedings of the IEEE International Conference on Computer Vision (ICCV)*, pages 2980–2988, 2017.
- [5] Shu Liu, Lu Qi, Haifang Qin, Jianping Shi, and Jiaya Jia. Path aggregation network for instance segmentation. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 8759–8768, 2018.
- [6] Wei Liu, Dragomir Anguelov, Dumitru Erhan, Christian Szegedy, Scott Reed, Cheng-Yang Fu, and Alexander C. Berg. SSD: Single shot multibox detector. In *European Conference on Computer Vision (ECCV)*, pages 21–37, 2016.
- [7] Joseph Redmon and Ali Farhadi. YOLO9000: Better, faster, stronger. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 7263–7271, 2017.
- [8] Joseph Redmon and Ali Farhadi. Yolov3: An incremental improvement. *arXiv preprint arXiv:1804.02767*, 2018.
- [9] Joseph Redmon, Santosh Divvala, Ross Girshick, and Ali Farhadi. You only look once: Unified, real-time object detection. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 779–788, 2016.
- [10] Shaoqing Ren, Kaiming He, Ross Girshick, and Jian Sun. Faster R-CNN: Towards real-time object detection with region proposal networks. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2015.
- [11] Sérgio Montazzolli Silva and Cláudio Rosito Jung. License plate detection and recognition in unconstrained scenarios. In *European Conference on Computer Vision (ECCV)*, pages 580–596, 2018.
- [12] Mingxing Tan, Ruoming Pang, and Quoc V. Le. Efficientdet: Scalable and efficient object detection. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 10781–10790, 2020.